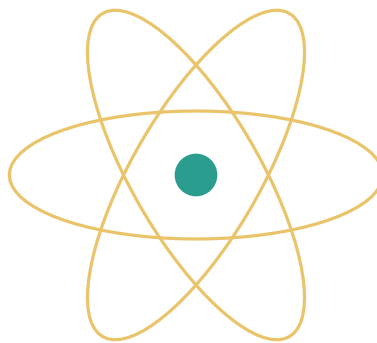




Nuclear Energy Educational Worksheets

Elementary (Grades 3–5) • Middle School (Grades 6–8)



Factual • Age-appropriate • Energy-source neutral





Nuclear Energy Word Search

Elementary School (Grades 3–5) • Word Search



ATOM BEACON

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Find the science words hidden in the puzzle. Words can go across, down, or diagonally. Circle each one as you find it.

S	B	L	O	D	S	A	F	E	T	Y	T
T	V	X	R	H	C	X	U	R	R	M	H
E	U	R	A	N	I	U	M	O	Q	R	Z
A	E	R	Z	R	R	Z	W	C	A	O	J
M	N	E	W	E	T	H	E	A	T	T	V
G	E	T	B	W	C	M	L	D	O	C	E
N	R	A	M	O	E	N	E	I	M	A	C
I	G	W	Z	P	L	V	U	R	K	E	N
L	Y	R	B	A	E	X	F	G	I	R	E
O	S	C	I	E	N	T	I	S	T	G	I
O	T	Y	O	E	N	I	B	R	U	T	C
C	H	R	E	E	N	I	G	N	E	U	S

WORD BANK

- ATOM
- REACTOR
- HEAT
- SAFETY
- COOLING
- ENGINEER
- ENERGY
- TURBINE
- SCIENCE
- WATER
- FUEL
- GRID
- STEAM
- POWER
- URANIUM
- ELECTRIC
- SCIENTIST
- CORE

DID YOU KNOW?

A nuclear power plant boils water to make steam. The steam spins a turbine, just like wind spins a pinwheel — and that turbine spins a generator that makes electricity!



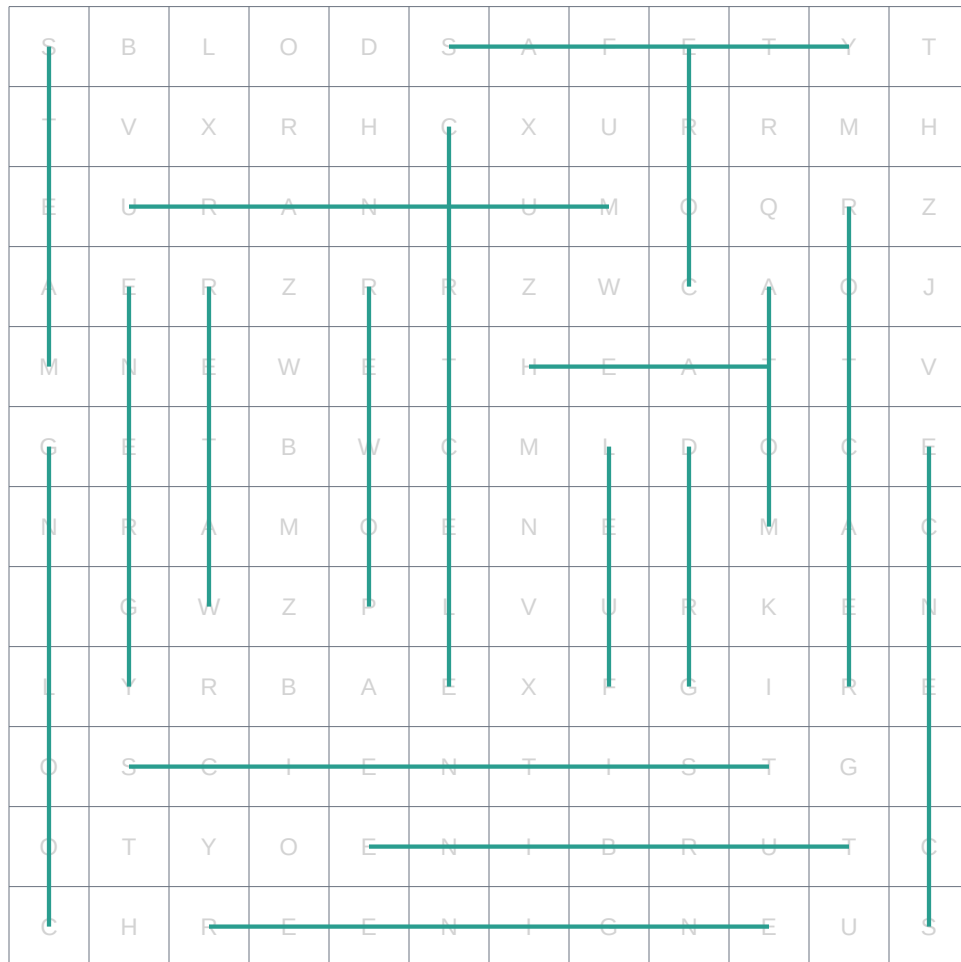


Nuclear Energy Word Search — Answer Key

Elementary • Word Search Key

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Power Plant Parts

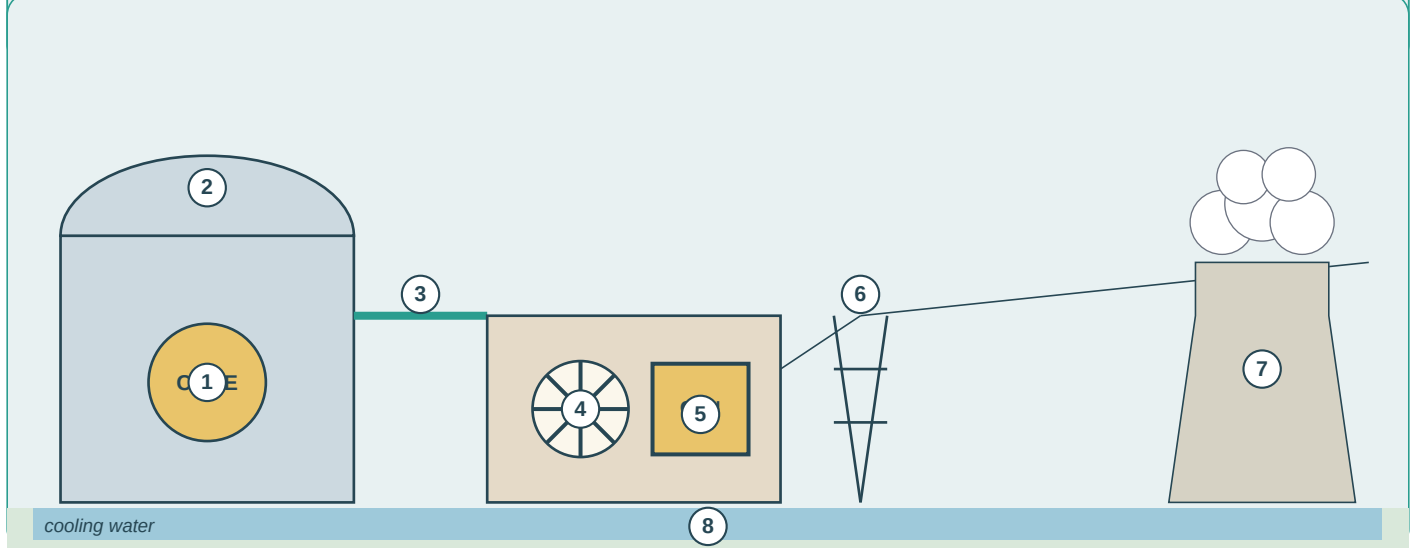
Elementary School (Grades 3–5) • Diagram Matching

ATOM BEACON

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Look at the parts of a nuclear power plant. Match each numbered part with the correct word from the word bank.



WORD BANK (match to numbers)

- | | |
|-----------------|------------------------|
| • Turbine | • Reactor |
| • Steam | • Containment Building |
| • Water | • Generator |
| • Cooling Tower | • Power Lines |

WRITE THE TERM NEXT TO ITS NUMBER

- | | |
|----------|----------|
| 1. _____ | 2. _____ |
| 3. _____ | 4. _____ |
| 5. _____ | 6. _____ |
| 7. _____ | 8. _____ |





Power Plant Parts — Answer Key

Elementary • Diagram Key



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Numbered labels:

1. Reactor
2. Containment Building
3. Steam
4. Turbine
5. Generator
6. Power Lines
7. Cooling Tower
8. Water





Connect the Dots: Cooling Tower

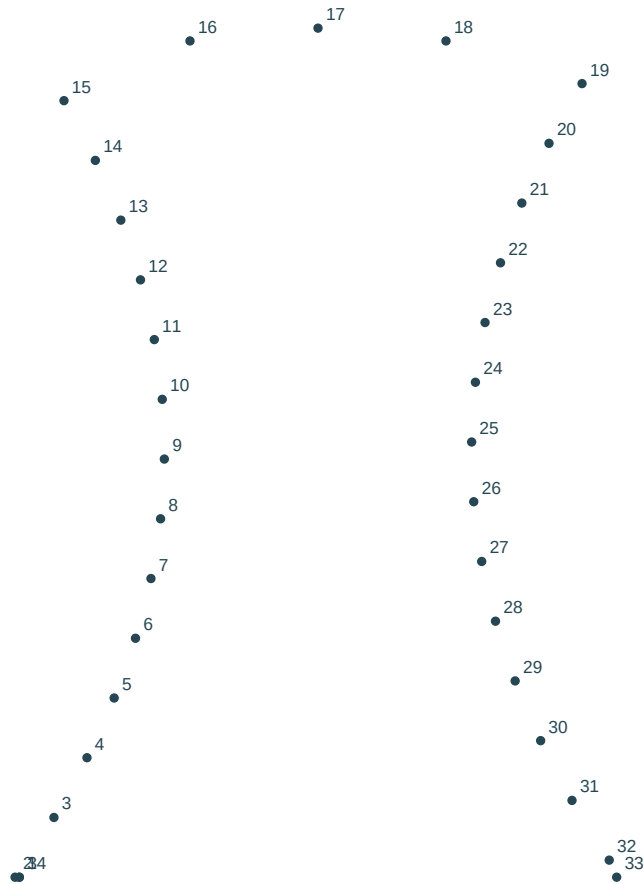
Elementary School (Grades 3–5) • Dot-to-Dot

ATOM BEACON

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Connect the dots in order from 1 to the end to reveal a famous shape you might see at a power plant.



AFTER YOU CONNECT THE DOTS:

You drew a COOLING TOWER. The white clouds you sometimes see rising from one are made of water vapor — just like steam from a tea kettle.

Color it in if you'd like! Fact: Cooling towers can be over 500 feet tall — taller than a 40-story building!



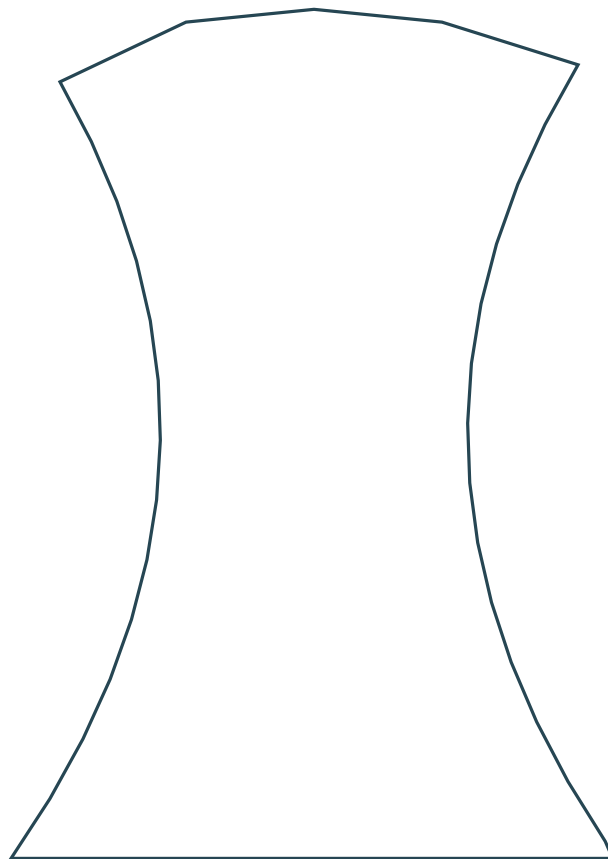


Connect the Dots: Cooling Tower — Answer Key

Elementary • Dot-to-Dot Key



ATOM BEACON
Nuclear Science Education





Reactor Systems Word Search

ATOM BEACON

Middle School (Grades 6–8) • Word Search

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Locate engineering vocabulary related to nuclear power generation. Words run horizontally, vertically, and diagonally — forward and backward.

T	U	R	B	I	N	E	V	L	E	Y	X	C	D	X	B
H	T	E	J	P	A	T	K	W	Z	F	U	E	L	D	J
O	R	M	P	N	R	A	N	R	T	H	E	R	M	A	L
G	A	O	K	E	N	E	F	A	E	G	R	X	R	G	G
N	D	D	R	U	O	R	S	I	L	A	O	Q	Y	I	R
I	I	E	S	T	I	L	O	S	S	O	C	P	T	X	U
D	A	R	T	R	S	E	O	T	U	S	O	T	W	W	R
L	T	A	E	O	S	G	V	R	A	R	I	C	O	A	A
E	I	T	A	N	I	R	S	M	T	R	I	O	S	R	N
I	O	O	M	E	M	I	V	A	Q	N	E	Z	N	B	I
H	N	R	I	P	S	D	F	B	F	J	O	N	E	G	U
S	Z	W	C	O	N	D	E	N	S	E	R	C	E	R	M
T	C	O	N	T	A	I	N	M	E	N	T	I	B	G	O
U	A	C	M	O	R	L	E	S	S	E	V	Y	V	I	V
V	L	D	A	S	T	J	U	M	N	I	A	H	C	H	M
Q	W	D	K	I	P	T	Y	L	B	M	E	S	S	A	R

WORD BANK

- CONTAINMENT
- COOLANT
- PRESSURIZER
- URANIUM
- SHIELDING
- THERMAL
- SAFETY
- MODERATOR
- GENERATOR
- CONDENSER
- REACTOR
- CHAIN
- GRID
- TURBINE
- RADIATION
- FISSION
- CORE
- ASSEMBLY
- TRANSMISSION
- ISOTOPE
- FUEL
- NEUTRON
- STEAM
- CONTROL
- VESSEL

DID YOU KNOW?

Most commercial reactors are pressurized water reactors (PWRs). Water in the primary loop stays liquid under high pressure, transferring heat to a separate secondary loop where steam is produced for the turbine.





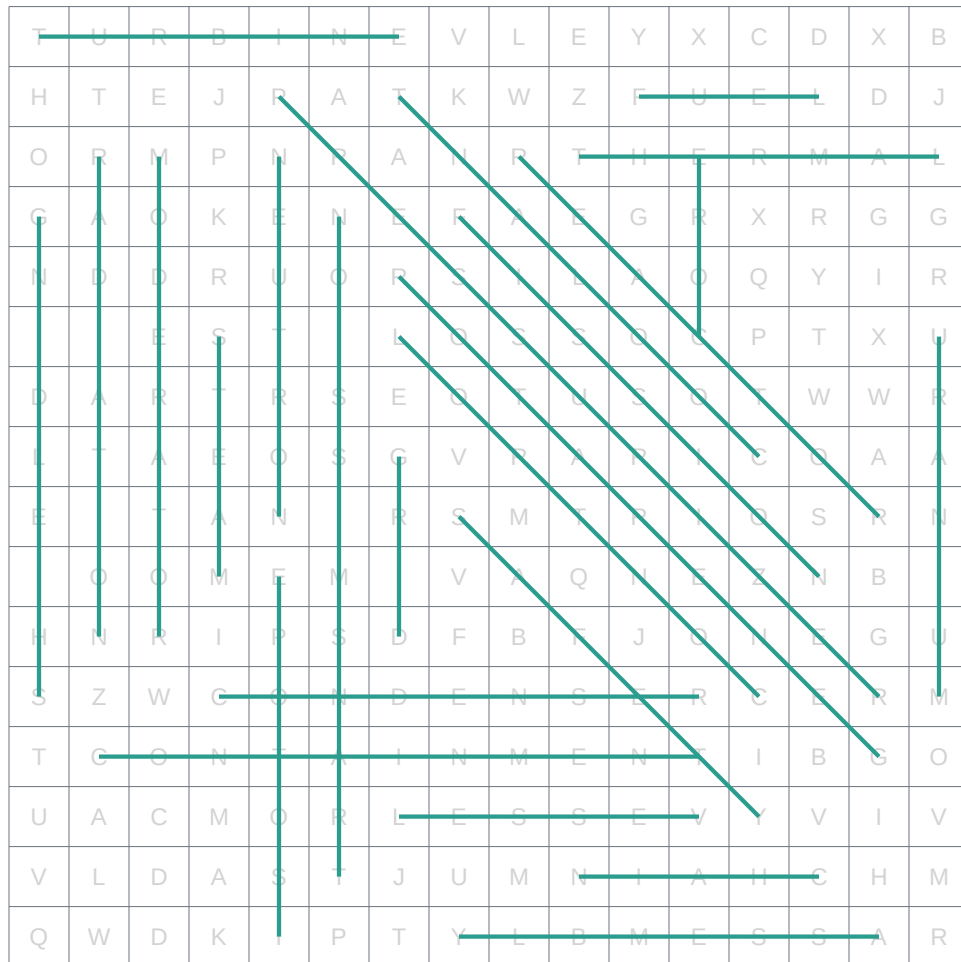
Reactor Systems Word Search — Answer Key

Middle School • Word Search Key



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Nuclear Science Education





Reactor Plant Cross-Section

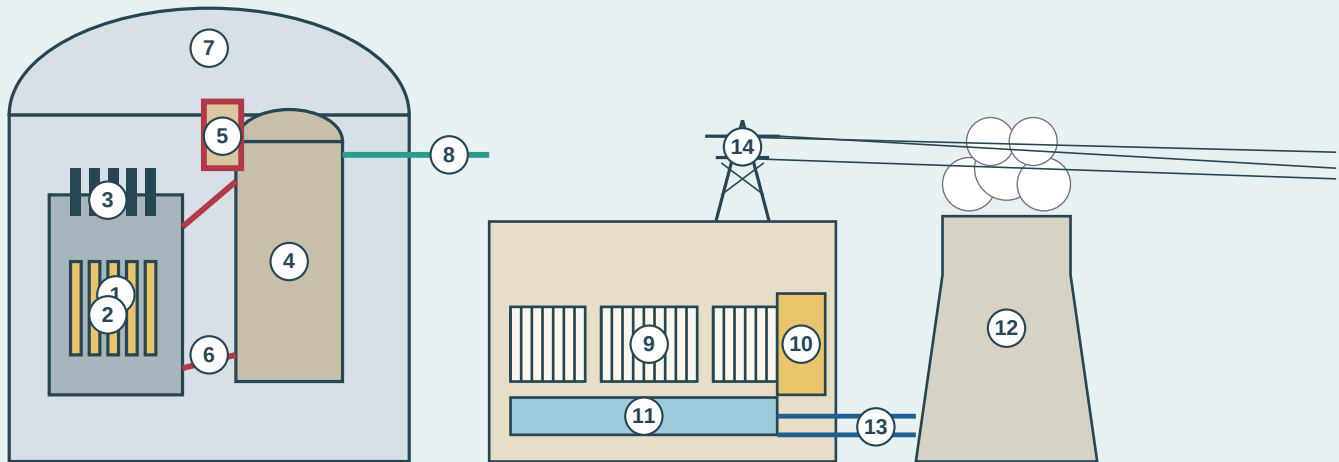
ATOM BEACON

Middle School (Grades 6–8) • Diagram Matching

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Identify the components of a pressurized water reactor (PWR). Match each numbered call out to a term in the



WORD BANK (match to numbers)

- Containment Structure
- Primary Coolant Loop
- Pressurizer
- Electrical Generator
- Condenser
- Steam Turbine
- Control Rods
- Cooling Tower
- Cooling Water Loop
- Transmission Lines
- Reactor Vessel
- Main Steam Line
- Steam Generator
- Fuel Rods

WRITE THE TERM NEXT TO ITS NUMBER

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____





Reactor Plant Cross-Section — Answer Key

Middle School • Diagram Key



ATOM BEACON

Nuclear Science Education

Numbered labels:

1. Reactor Vessel
2. Fuel Rods
3. Control Rods
4. Steam Generator
5. Pressurizer
6. Primary Coolant Loop
7. Containment Structure
8. Main Steam Line
9. Steam Turbine
10. Electrical Generator
11. Condenser
12. Cooling Tower
13. Cooling Water Loop
14. Transmission Lines





Connect the Dots: Reactor → Turbine → Grid

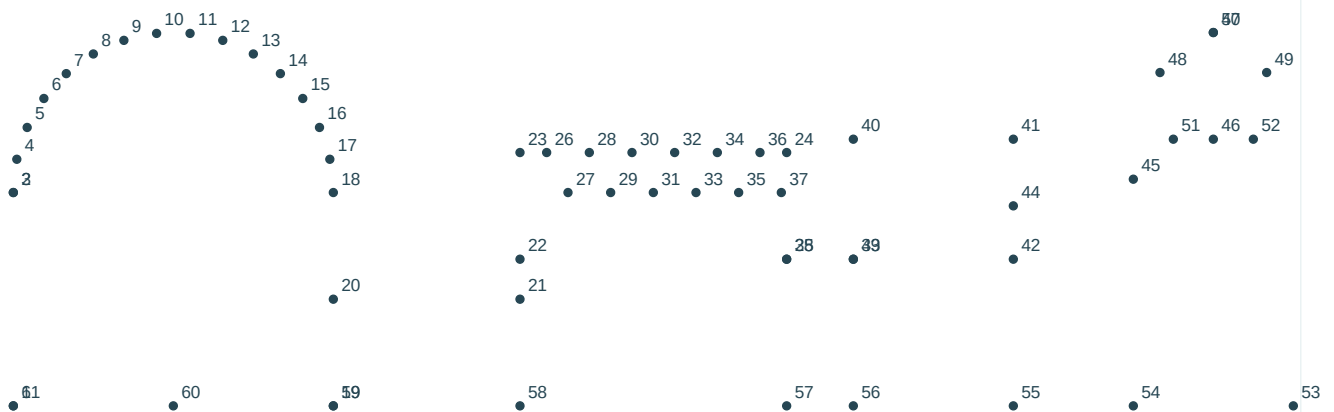
ATOM BEACON

Middle School (Grades 6–8) • Dot-to-Dot

Nuclear Science Education

Name: _____ Class: _____ Date: _____

Connect the dots from 1 to the end to reveal the energy path from a reactor through a turbine and generator out to the electrical grid.



AFTER YOU CONNECT THE DOTS:

You drew the basic energy-conversion chain of a power plant: thermal energy → mechanical energy → electrical energy.

Color it in if you'd like! *Fact: Generators at large power plants can produce more than 1,000 megawatts — enough electricity for hundreds of thousands of homes.*

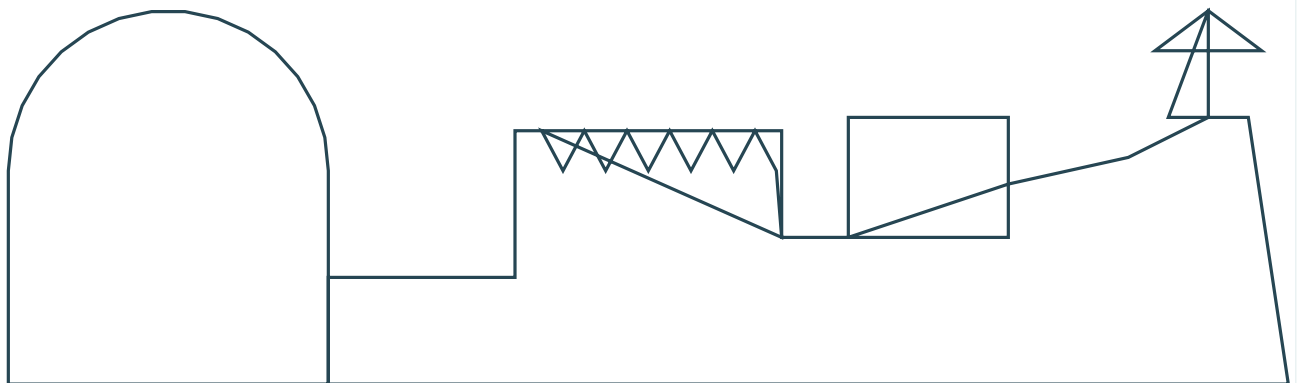




Connect the Dots: Reactor → Turbine → Grid — Answer Key

Middle School • Dot-to-Dot Key

BEACON
Nuclear Science Education





Teacher Packet — Educator Notes

Vocabulary glossary • Discussion • STEM alignment



ATOM BEACON

Nuclear Science Education

About this packet. These worksheets introduce nuclear energy technology with a neutral, science-first tone. They focus on how reactors generate electricity, the role of safety systems, and the vocabulary engineers use. They do not advocate for or against any energy source.

Vocabulary glossary.

Atom — smallest unit of an element. **Fission** — splitting of a heavy nucleus, releasing heat. **Uranium** — common nuclear fuel element. **Fuel pellet / Fuel rod / Fuel assembly** — ceramic pellets stacked in metal tubes, bundled into assemblies. **Reactor core** — region where fuel and control rods sit. **Control rod** — neutron-absorbing rod that adjusts the chain reaction. **Moderator** — material (often water) that slows neutrons so fission continues. **Coolant** — fluid that carries heat out of the core. **Reactor vessel** — heavy steel container holding the core. **Pressurizer** — keeps primary coolant under high pressure. **Steam generator** — heat exchanger that boils secondary water into steam. **Turbine** — bladed wheel spun by steam. **Generator** — converts spinning motion into electricity. **Condenser** — cools used steam back into water. **Cooling tower** — releases waste heat to the air. **Containment structure** — reinforced building enclosing the reactor. **Radiation shielding** — concrete, water, or lead that blocks radiation. **Isotope** — atoms of the same element with different neutron counts. **Chain reaction** — self-sustaining series of fission events. **Transmission lines** — high-voltage wires moving electricity to the grid.

Discussion questions.

1. How is a nuclear power plant similar to other steam-driven power plants?
2. Why are several safety barriers used between the fuel and the outside environment?
3. What jobs do scientists and engineers do at a power plant?
4. Why is heat such an important part of generating electricity?
5. What does "shielding" mean and where do we see it in everyday life (e.g., dental x-ray aprons)?

Suggested NGSS / STEM alignment.

Elementary: 4-PS3 Energy; 4-ESS3-1 Energy & Resources.

Middle School: MS-PS1-1 Atomic structure; MS-PS3-3/5 Energy transfer; MS-ESS3-4 Human impacts of resource use; MS-ETS1 Engineering design.

Educational neutrality disclaimer. These materials present nuclear technology factually. They are not intended to promote or discourage any energy source. All energy sources—coal, oil, natural gas, hydro, geothermal, solar, wind, and nuclear—have roles in modern energy systems and should be discussed respectfully.

Print tips. Letter size, single-sided recommended for puzzles. Grayscale-safe. Each worksheet is self-contained; mix and match by grade level.

